
CE 354P: Building and Pavement Materials Laboratory

Course Code	CE 354P
Course Name	Building and Pavement Materials Laboratory
L-T-P-C	0-0-2-1
Prerequisites	IC 140 - Graphics for Design
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)

Laboratory Modules

Environment

- Determination of specific gravity, fineness and soundness of different cements.
- The study of setting and hardening characteristics of different cements.
- Determination of specific gravity, moisture content, bulking and water absorption of aggregates.
- The study of abrasion, attrition, hardness, shape, grading and packing characteristics of aggregates.
- The study of water absorption, sorptivity and permeability characteristics of concrete and brick samples.
- Assessing the presence of organic impurities and fines in aggregates.
- Assessment of surface hardness, flexural strength, compressive strength and stress-strain characteristics of cement paste, mortar, concrete and clay & fly-ash bricks.
- Testing of filling ability and passing ability of self compacting concrete.
- The study of stress-strain characteristics of reinforcement bars, other metals and alloys.
- The study of consistency and ductility of bitumen samples.
- Determination of flash and fire points of bitumen samples.
- Determination of softening point of bitumen samples.
- Determination of viscosity of bitumen samples.
- To determine the optimum binder content for a bituminous mix by Marshall's method.